

Kubernetes: Open source Kubernetes is free and downloaded from its repository on GitHub. Administrators must build and deploy the Kubernetes release to a local system or cluster, or to a system or cluster in a public cloud such as AWS, Google Cloud Platform (GCP) or Microsoft Azure.

Puppet: This is a cloud DevSecOps tool for operating and delivering software. Puppet automates deployment to provide reliability and agility. It provides continuous automation and delivery faster across the complete software delivery life cycle. Also, the tool increases productivity and operational efficiency, infrastructure as code, configuration management, automated testing and continuous delivery.

Veracode: This powerful cloud based service suite for software testing can help to implement end-to-end security. It provides application security services and solutions to reduce risk in Web, mobile and third party applications. The various security services provided by Veracode for DevSecOps are:

- Static analysis security testing
- Software composition analysis
- Vendor analysis security testing
- Web application scanning

Selenium: This is an automated functional testing tool to test Web applications. Installed as a Firefox browser plugin, it helps to record and playback test scenarios. In a DevSecOps scenario, once the application is deployed in a test environment, Selenium automated testing is invoked.

Supergiant: This open source platform for container management can be utilised for Kubernetes deployment on multiple clouds in a matter of minutes. The Supergiant API is used for streamlining production deployment.

Apache Mesos: Apache Mesos abstracts CPU, memory, storage and other computer resources away from machines, whether they are physical or virtual, and enables fault-tolerant and elastic distributed systems to be easily built and run effectively. The Mesos kernel runs on every machine and provides applications like Hadoop, Spark, Kafka and Elasticsearch, with APIs for resource management and scheduling across the entire data centre and cloud environments.

Synk: This open source security management tool is used to automatically find, prioritise and fix vulnerabilities in the open source code and its dependencies. It helps in developing cloud native applications.

Benefits of DevSecOps

Some of the key benefits of adopting DevSecOps processes are:

- It eliminates silos, and promotes collaboration and team work.
- It identifies the vulnerabilities, and reduces the cost and time to deliver software.
- A DevSecOps tools setup reduces the time of deployment by 80-90 per cent. As an example, it reduces deployment time from 12 hours to 2 hours.
- It increases software quality with automated testing. It also reduces the cost and time needed to test, and deployment related downtime.
- It provides improved and stable operations, diminishes security threats, reduces rework, and increases the reliability of service delivery.
- It improves development productivity and overall software quality by 20 per cent with automated and early detection of defects in the cycle.
- Improves business value and provides increased customer value by being responsive to change.

To sum up, the combination of cloud computing and DevSecOps provides automated and fast application development and monitoring. This increases an organisation's ability to deliver applications and services at high velocity. **END** 

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